An Adaptive Network for Enhanced Vision and Semantic Cooperative Understanding in Aerial Object Detection

Xin Cheng, Yibo Liu, Haisu Zhang, Huan Liu, Yao Wu, Zhiliang Hu, and Ning Xu, *Member, IEEE*

***Abstract*—Object detection in remote sensing imagery faces challenges such as extreme scale variations and complex backgrounds. Although current methods have made significant strides in visual feature extraction, their predominant focus remains on the image itself, overlooking the potential of integrating external knowledge. To address this limitation, we introduce the knowledge-aware network with region-adaptive fusion for detection (KARFDet), which seamlessly integrates region-specific semantic information with visual features. First, a multiscale fused kernel attention (MSFKA) module is introduced, leveraging a parallel multibranch architecture to enhance contextual feature extraction. Second, a knowledge graph semantic extraction (KGSE) module is designed, employing the random walk with restart (RWR) algorithm to transform discrete knowledge into computable semantic associations. Finally, a novel triple-order knowledge integration (TOKI) mechanism is proposed, which adaptively fuses original, second-order, and probabilistic semantic knowledge, dynamically allocating knowledge weights based on target scale characteristics. Experiments on the DIOR, NWPU VHR-10, and SIMD datasets show that KARFDet achieves mAP_{50} scores of 65.6%, 91.9%, and 75.8%, respectively, significantly outperforming the baseline model and establishing a new paradigm for semantic-aware detection in complex scenarios. The code is available at https://github.com/ChengXCode/KARFDet**

***Index Terms*—Knowledge graph, object detection, remote sensing, semantic enhancement.**

I. Introduction

THE rapid advancement of high-resolution remote sensing has greatly improved Earth observation, making object detection essential in areas such as emergency response, resource exploration, and land monitoring [1]. However, the distinct characteristics of remote sensing imagery present major challenges to standard detection models [2]. First, the substantial scale variations in aerial imagery, ranging from hundred-meter-long runways to meter-sized vehicles, pose a significant challenge to convolutional networks with inherently limited receptive fields [3], [4]. Second, small objects like ships or cars may occupy only dozens of pixels when captured from high altitudes, making their features easily lost in complex backgrounds [5], [6]. Additionally, occlusion and overlap are common in remote sensing scenes; vehicles in cities often obscure each other, causing fragmented features, while stacked containers in ports show strong visual adhesion, lowering separation accuracy [7]. More importantly, remote sensing scenes involve complex spatial semantic relationships—for example, airports typically include airplanes, and ports involve ships. Such logic cannot be fully learned through pixel-level features. When objects are heavily occluded or spectrally confused, visual features lose discriminability, causing incomplete contours, misaligned boxes, and missed detections [8]. Ambiguous features may also mislead the model into producing false positives [9]. Approaches relying solely on visual features struggle to fundamentally overcome these errors.

Recent mainstream remote sensing detection methods also show clear limitations. Traditional detectors like Faster R-CNN [10] and earlier versions of the YOLO family [11], [12], [13], [14] rely heavily on hand-crafted anchor boxes, limiting their adaptability to extreme object scale distributions in remote sensing [15], [16]. Although RT-DETR [17], as a real-time detection transformer, eliminates the need for hand-crafted anchors through its end-to-end set prediction mechanism, its uniform feature processing still falls short in handling extreme scale variations. Its fixed receptive field struggles to capture both global layouts and local details simultaneously. More critically, these methods remain confined to data-driven, pixel-level feature learning, which is essentially a form of visual perception. When targets are severely occluded, extremely small, or spectrally confused, these purely visual features become unreliable, preventing models from reasoning in accordance with scene logic.

Semantic associations among geographic entities form the basis for knowledge enhancement. As shown in Fig. 1, models relying solely on pixel-level features for isolated object detection disregard external knowledge value, resulting in fragmented semantics. Such models cannot comprehend semantically legitimate associations like ships gathering in harbors (HBs) or airplanes at airports, nor identify objectively implausible combinations such as ships beside stadiums or

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Xin Cheng, Huan Liu, Zhiliang Hu, and Ning Xu are with the School of Information Engineering, Wuhan University of Technology, Wuhan 430070, China (e-mail: chengcx@whut.edu.cn; ballrant@whut.edu.cn; hzliang@whut.edu.cn; xuning@whut.edu.cn).

Yibo Liu, Haisu Zhang, and Yao Wu are with the Information Support Force Engineering University, Wuhan 430000, China, and also with Hubei Provincial Key Laboratory of Data Intelligence, Wuhan 430014, China (e-mail: liuyibo@nudt.edu.cn; zhanghaisu@nudt.edu.cn; yaowu9507@163.com).

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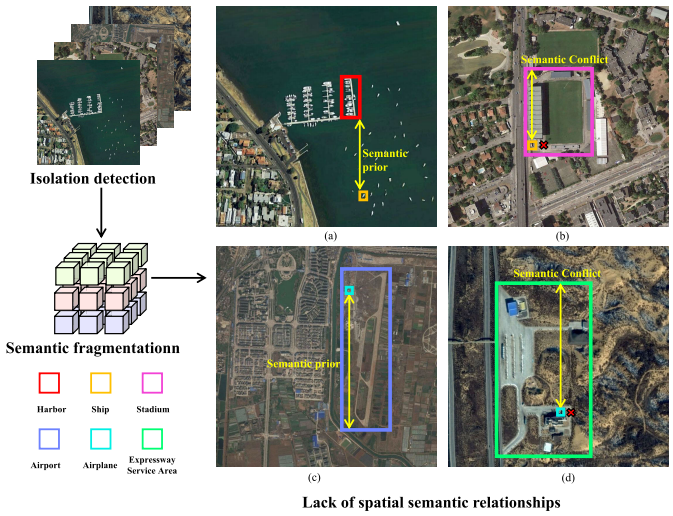


Fig. 1. Isolated detection models lack semantic knowledge about real-world objects. This leads to their failure in identifying and improving illogical object detection results, such as a ship in a stadium or an airplane on an expressway service area. (a) HB and ship. (b) Stadium and ship. (c) Airport and airplane. (d) Expressway service area and airplane.

airplanes at highway service areas, revealing their fragmented and constrained internal knowledge representation.

To address these limitations, we propose a knowledge-aware network with region-adaptive fusion for detection (KARFDet), a framework bridging visual perception with knowledge graph semantic structures to achieve cognitive reasoning. It encodes semantic relationships and spatial priors in structured form, injecting explicit domain knowledge into implicit deep learning models to ensure semantic prediction consistency. In visually challenging scenarios, our method adaptively leverages knowledge according to target scale, enabling a cognitive leap beyond pure perceptual detection. This advances remote sensing object detection from merely seeing objects to genuinely understanding scenes, providing an extensible and innovative solution for multisource information fusion. KARFDet does not replace visual recognition with external knowledge. Instead, it performs structured semantic calibration on the initial predictions produced by the visual detector. KARFDet is mainly designed for remote sensing object detection under relatively homogeneous sensing conditions, and its applicability to cross-sensor scenarios such as optical-to-infrared transfer remains beyond the primary scope of this study.

The main contributions of this work are summarized as follows.

1) We propose KARFDet, which, to the best of our knowledge, is the first framework to synergistically couple RT-DETR with knowledge reasoning by spatially encoding a computable visual-symbolic mapping, forming a cognitive closed loop for bidirectional enhancement.
2) We design the multiscale fused kernel attention (MSFKA) module to enhance the RT-DETR backbone's adaptability to extreme scale variations. It employs parallel multiscale separable convolutional kernels to capture rich contexts and generates spatial attention weights through adaptive fusion.
3) We introduce a knowledge graph semantic extraction (KGSE) module using the random walk with restart (RWR) algorithm to quantify intercategory association strength. This transforms discrete symbolic knowledge into a continuous, computable semantic association matrix, providing structured priors for knowledge fusion.
4) We propose the triple-order knowledge integration (TOKI) module for semantic knowledge fusion, which adaptively adjusts weights among first-order knowledge, second-order reasoning, and normalized knowledge based on proposal area, enabling precise semantic calibration and suppressing inconsistent false detections.

As a follow-up to our prior work KGVA-YOLO [18], which served as an initial exploration of knowledge graph-enhanced remote sensing object detection by adopting YOLOv8 with a multiscale regional interactive attention (MRIA) module and a static semantic enhancement module, KARFDet advances this research line through three major architectural innovations as follows.

1) It upgrades the backbone from YOLOv8 to RT-DETR, whose transformer-based architecture with dynamic queries enables superior handling of extreme scale variations.
2) It replaces the MRIA module with the proposed MSFKA module, which leverages a parallel multibranch design for richer multigranularity feature extraction.
3) It introduces the TOKI mechanism, which dynamically and adaptively fuses first-order, second-order, and normalized semantic knowledge, overcoming the limitations of the static fusion strategy in KGVA-YOLO.

II. Related Works

A. Object Detection Algorithms

Object detection, a core computer vision task, has evolved from traditional handcrafted feature methods to end-to-end deep learning models [19]. Early sliding-window approaches with handcrafted features struggled with limited representation in remote sensing imagery, failing to handle object diversity in complex backgrounds [20]. Deep learning spurred two-stage detectors like the R-CNN series and single-stage models such as YOLO family [11], [12], [13], [14], [21], [22], [23], [24], [25], [26], [27] and SSD [28]. R-CNN used selective search for region proposals followed by CNN feature extraction, albeit inefficiently [29]. Fast R-CNN [30] introduced RoI pooling for feature sharing, while Faster R-CNN [10] achieved end-to-end training via a region proposal network (RPN), significantly accelerating detection. Single-stage methods like YOLO frame detection as regression, predicting object locations and categories directly in one forward pass—achieving speed advantages with slight accuracy trade-offs. RetinaNet [31] balanced this with focal loss to address class imbalance.

For remote sensing challenges, specialized improvements have emerged. LSKNet [32] dynamically adjusts convolutional kernels and employs large selective kernels to capture long-range context, excelling in complex scenes. For few-shot object detection in remote sensing, GE-FSOD [33] enhances generalization capability by introducing a cross-level fusion pyramid attention network, a multistage refinement

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RPN, and a generalized classification loss. SuperYOLO [34] integrates super-resolution branches to enhance feature resolution under low-input quality and uses pixel-level fusion to reduce computational redundancy, improving detail representation in multimodal imagery. The SOAR algorithm [35] combines Vision Mamba with bidirectional state space modeling, alleviating small object occlusion by background noise and boosting recognition stability in complex environments. These approaches advance remote sensing detection accuracy through multiscale feature fusion and attention mechanisms.

B. DETR-Based Object Detection Algorithms

RT-DETR [17], proposed by Baidu, introduces a core innovation through its efficient hybrid encoder design. By decoupling intrascale interaction and cross-scale fusion via the AIFI and CCFM modules, respectively, it significantly boosts efficiency while maintaining high accuracy, outperforming many contemporary real-time detectors. Inspired by such architectural advancements, a series of studies has been dedicated to adapting and enhancing the DETR [36] framework for specialized remote sensing scenarios. For instance, MMR-DETR [37] addresses road damage detection under complex backgrounds by incorporating multiscale self-attention and cross-fusion modules. Similarly, MCIF-DETR [38] tackles insulator fault detection with a directional feature enhancement module, while MD-DETR [39] mitigates SAR image noise through dedicated denoising and guided learning mechanisms. Further extending this trend, KDFF-DETR [40] enhances spaceborne infrared ship detection using Fourier context encoding. The CSFPR-RTDETR [41] model employs an innovative combination of spatial-frequency domain analysis and geometric priors, thereby substantially improving the accuracy of real-time small object detection in UAV-acquired images.

Despite their specialized advancements, the aforementioned DETR variants primarily focus on enhancing visual feature representation, often overlooking the rich semantic relationships and structured prior knowledge inherent in remote sensing scenes.

C. Knowledge Graphs

Deep learning methods for remote sensing object detection face inherent limitations, including heavy reliance on annotated data, limited interpretability, and vulnerability to noise and attacks—making them unsuitable for high-reliability scenarios [42]. Knowledge graphs address these issues by structuring domain knowledge using entity–relation–attribute triplets, which effectively capture semantic and spatial relationships in geospatial contexts [43], [44]. By organizing fragmented information into interconnected networks, Knowledge graphs surpass traditional data representations in expressiveness and intuitiveness [45].

Building upon this foundational advantage, a series of studies has demonstrated the versatile application of KGs across various remote sensing tasks. To address the different categories with the same spectral issue in remote sensing, GSAN [46] constructs a geographical knowledge graph and employs graph structure attention to enhance semantic discrimination by modeling spatial context. ARNet [47] introduces a knowledge graph for fine-grained aircraft recognition, utilizing spatial context and component features to address complex backgrounds and class imbalance in remote sensing imagery. For remote sensing scene classification, Evo RSFormer [48] introduces a hybrid architecture that synergistically combines evolutionary CNN with Transformer, and employs a knowledge-guided neural architecture search to optimize the network design. KGVA-YOLO [18] serves as an initial effort to integrate knowledge graphs with the YOLO architecture, demonstrating the potential of semantic knowledge in remote sensing object detection. By incorporating a knowledge graph-based semantic enhancement module into the YOLO framework, KGVA-YOLO validates that external structured knowledge can effectively assist visual features in recognizing objects under challenging conditions such as occlusion and scale variation. However, its relatively simple semantic fusion strategy and the inherent limitations of the YOLO-based detector leave room for further improvement in handling extreme scale diversity and complex spatial reasoning.

In conclusion, the integration of knowledge graphs marks a paradigm shift in remote sensing image analysis. It transitions models from purely data-driven learning to a collaborative knowledge-data framework, thereby establishing a more reliable and interpretable foundation for advanced image understanding. Knowledge graphs can be categorized by domain into open-domain knowledge graphs and vertical-domain knowledge graphs. Open-domain knowledge graphs target the entire Internet user base, covering a wide range of semantic knowledge. They are constructed through large-scale automated data extraction and integration, providing semantic support for various application scenarios. Notable existing open-domain knowledge graphs include Google's Freebase [49], ConceptNet [50], DBpedia [51], OpenIE [52], and Wikipedia's WikiData [53]. Among these, ConceptNet is an open-source knowledge graph that evolved from the crowdsourced project Open Mind Common Sense, launched in 1999 at the MIT Media Lab. It has since grown to incorporate knowledge from other crowdsourced resources, with Wiktionary being its largest input source. ConceptNet comprises over 21 million edges and 8 million nodes, making it a robust foundation for the research and experiments in this article.

III. Proposed Method

A. Overview

As illustrated in Fig. 2, KARFDet's visual backbone is based on RT-DETR [17], utilizing HGNetv2 to extract multiscale features. The neck employs a hybrid FPN and PAN structure for feature fusion. A key enhancement is the novel MSFKA module embedded at three critical points postfusion. Using parallel multibranch convolutions, MSFKA captures multigranularity context, significantly boosting representation for extreme scale variations and providing robust visual proposals. In parallel, the KGSE module processes a knowledge

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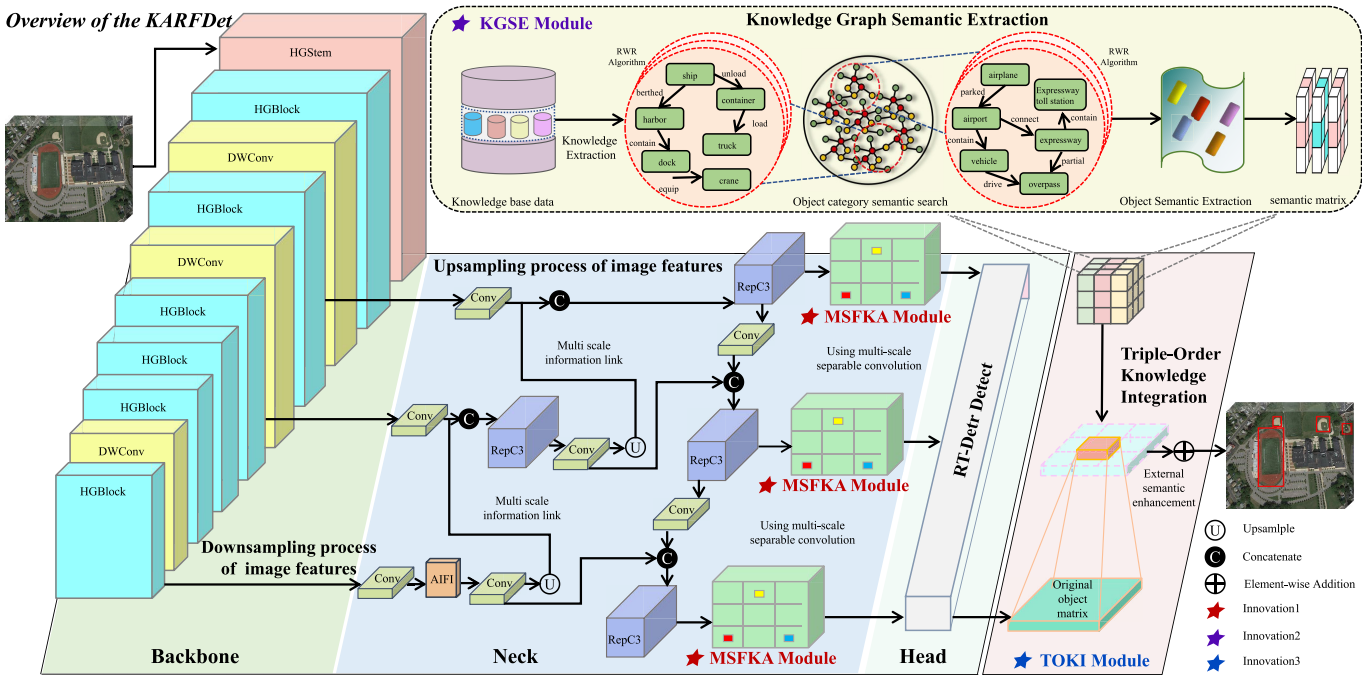


Fig. 2. Architecture of KARFDet highlights three key innovative modules. 1) A parallel multibranch structure that enhances multiscale feature representation capability. 2) Domain adaptation of the knowledge graph for remote sensing, generating a semantic association matrix. 3) A dynamic knowledge fusion mechanism based on target uncertainty, enabling adaptive calibration to suppress false detections.

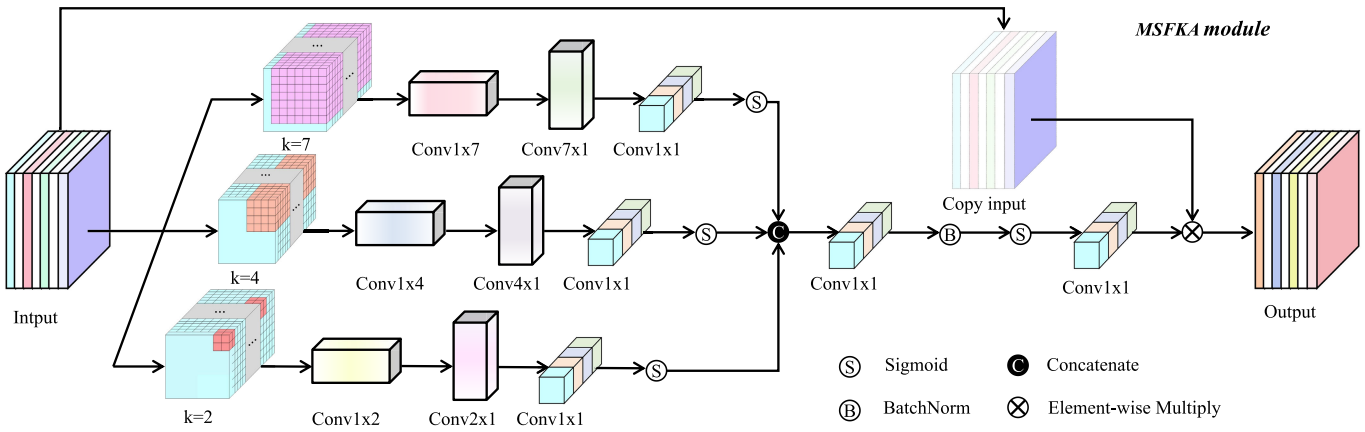


Fig. 3. MSFKA module generates spatial attention via parallel multiscale fusion, simultaneously capturing global contexts and local details to robustly represent extreme object scales in remote sensing.

graph via the RWR algorithm to generate a semantic association matrix, which quantifies the relational strength between object categories. The innovative TOKI mechanism then uses this matrix to dynamically calibrate initial classifications. Fusion employs an adaptive weighting mechanism based on each target's visual confidence, proxied by its area percentile. Large, clear targets rely more on first-order ontological knowledge, while ambiguous small targets depend more on second-order contextual reasoning and normalized knowledge. This achieves precise semantic calibration. The fusion process in KARFDet follows a staged decision pathway. The visual branch first produces proposal features and initial classification logits, and the TOKI module then refines these visually derived predictions using explicit semantic relations encoded in the knowledge graph.

In summary, KARFDet establishes a synergistic architecture where the enhanced visual backbone handles localization and the knowledge graph performs contextual validation and semantic calibration, working together to improve interpretation accuracy and robustness.

B. MSFKA Module

As illustrated in Fig. 3, the core idea of the MSFKA module is to leverage multiple parallel convolutional branches to extract features at varying scales, fuse these features to generate a spatially and channel-wise adaptive attention weight map, and finally calibrate the input features through an element-wise multiplication.

The MSFKA module first takes the feature map X from the previous layer as input, where $X \in \mathbb{R}^{B\times C\times H\times W}$, with B being

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the batch size, C the number of channels, H the height, and W the width of the feature map. We first retain a copy of the input feature map, denoted as U, to preserve the original information for subsequent fusion.

Three parallel branches process the input feature map, each employing large, medium, and small kernel scales, respectively. Each branch sequentially performs horizontal and vertical feature aggregation using depthwise separable convolutions with $1 \times k$ and $k \times 1$ kernels, followed by channel compression via a standard 1×1 convolution and SiLU activation, producing multiscale feature representations Y_i. The operation of branch i with kernel size k_i is formally defined as

$$Y_i = \text{SiLU}\left(\text{Conv}_{1\times 1}\left(\text{Conv}_{k_i\times 1}\left(\text{Conv}_{1\times k_i}(X)\right)\right)\right) \tag{1}$$

here, both $\text{Conv}_{1\times k_i}$ and $\text{Conv}_{k_i\times 1}$ represent depthwise separable convolutions, $\text{Conv}_{1\times 1}$ represents standard convolution, and SiLU indicates the activation function.

The outputs of the three branches are concatenated along the channel dimension to form a tensor Y

$$Y = \text{Concat}(Y_1, Y_2, Y_3) \tag{2}$$

where Concat denotes concatenation along the channel axis, resulting in a tensor of size $B \times 3C \times H \times W$, i.e., three times the original channel number.

This concatenated feature is then fused through a 1×1 convolution followed by batch normalization and a Sigmoid activation, which constrains values to $[0, 1]$ and establishes a weighting basis for subsequent attention, producing the preliminary feature map F

$$F = \text{Sigmoid}\left(\text{BN}\left(\text{Conv}_{1\times 1}(Y)\right)\right). \tag{3}$$

To further enhance the expressiveness of the attention mechanism, an additional 1×1 convolution is applied to F, refining the attention weights without altering the dimensions. This yields the final attention weight tensor $\mathbf{A}$

$$\mathbf{A} = \text{Conv}_{1\times 1}(\mathbf{F}). \tag{4}$$

The purpose of the above convolution is to perform nonlinear refinement and channel recalibration on feature weights. While F contains multiscale information, the preceding Sigmoid function essentially squashes the fused features into a static, per-position weighting map. The representational power and adaptability of this map might be limited. Applying an additional, standalone 1×1 convolution on F can serve as a lightweight, learnable nonlinear transformation. It performs this refinement without changing the spatial or channel dimensions of the attention map.

Finally, the output feature map $\mathbf{O}$ is obtained by performing an element-wise multiplication between the original input reference U and the attention weight $\mathbf{A}$

$$\mathbf{O} = \mathbf{U} \odot \mathbf{A} \tag{5}$$

where $\odot$ denotes element-wise multiplication, ensuring $\mathbf{O}$ maintains the same dimensions as $\mathbf{X}$.

This multiscale integration enhances sensitivity to objects while suppressing background noise in remote sensing imagery.

C. KGSE Module

To incorporate high-level semantic reasoning, we integrate an external knowledge graph that enhances contextual awareness through structured knowledge. As shown in Fig. 2, this module quantifies intercategory semantic relationships from open-domain knowledge into a semantic consistency matrix, which serves as prior knowledge to semantically validate and optimize the initial class logits generated by the visual model in subsequent fusion modules.

This article introduces an external large-scale knowledge graph to enhance the semantics of object detection tasks. First, the original content of the knowledge graph is preliminarily processed to generate a list of all triplets, which are then processed into a form called graph Gra_{base} that can be calculated by the RWR algorithm for semantic extraction. Each edge is represented as a triplet <Node index, Target node index, Weight>. Among them, the node index is the unique ID of the concept in ConceptNet, and the weight is the confidence score of the relationship in ConceptNet, which serves as the edge weight of the random walk. At the same time, generate a lookup table between node indexes and concept names for subsequent queries. The subsequent detailed process can be found in Algorithm 1.

Algorithm 1 Knowledge Graph Processing

Input: List of Detection Categories $List_C$, knowledge graph Data Gra_{base}
Output: Semantic matrix h

```
Initialize RWR on graph Gra_base for each category i in List_C do
if node i exists in Gra_base then
    for each category j in List_C do
        if node j exists in Gra_base then
            Compute RWR score R_{i,j} from node i to j: R_{i,j} = lim_{t→∞} P(v_j | v_i; a)
        end if
    end for
end if
end for
for each pair of categories i, j in List_C do
    Compute bidirectional semantic distance: s[i, j] = s[j, i] = ∛(R_{i,j} × R_{j,i})
end for
return Semantic matrix s
```

The knowledge graph is formally defined as a directed graph

$$G = (V, E, R) \tag{6}$$

where $V = \{v_1, v_2, \ldots, v_N\}$ denotes the entity set, directly corresponding to target categories in our detection taxonomy. $R = \{r_1, r_2, \ldots, r_K\}$ represents the relation type set, including proximity, part-of, functional association, etc. The set of edges $E \subseteq V \times R \times V$ indicates the collection of factual relationship triplets.

For a specific dataset, we define its M categories as

$$L = \{C_1, C_2, \ldots, C_M\}. \tag{7}$$

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Entity linking is then applied to match these categories with corresponding knowledge graph entities, from which we extract a multihop neighborhood subgraph G_{sub} to construct a task-focused, context-rich semantic network.

When linking entities, to ensure accuracy, a precise matching strategy is adopted by first using the official category names of the dataset as seed concepts. After removing hyphens from category labels and converting them to lowercase, we directly use traversal algorithms to find nodes in ConceptNet that are identical to these concepts, ensuring the accuracy of semantic alignment. For example, "storage tank" is mapped to the standard concept "storage_tank" in ConceptNet. With a sufficiently large knowledge base, this approach can comprehensively discover all target concepts and ensure accuracy.

The selection of entities and relationships is closely related to the knowledge background noise. ConceptNet contains a large number of relationships and entities, some of which are important for understanding the semantics of the target, while inappropriate entity and relationship selection may introduce noise in the knowledge. The selection of entities is quickly obtained through the algorithm of entity linking mentioned above. For the screening of relationship R, it is necessary to focus on the effectiveness of the relationship in connecting target category nodes. The target association strength (TAS) is defined as follows:

$$\text{TAS}(r) = \frac{|E_r(L, L)|}{|E_r|}. \tag{8}$$

E_r is all edges of type r, where $E_r(L, L)$ is the edge connecting two target category nodes. The higher the TAS ratio, the more effective the relationship is in directly associating target categories. Ultimately, the best set of relationship categories is selected based on the ranking of TAS scores. To quantify the semantic association strength between any two categories C_i and C_j in subgraph G_{sub}, we employ the RWR algorithm. This simulates a random walk starting from a source node: at each step, with probability $1 - \alpha$ the walker moves to a neighbor according to the transition matrix P, or with probability α restarts from the source. This mechanism effectively captures both direct and indirect multihop semantic relationships. The restart probability $\alpha \in (0, 1)$ controls the locality–globality trade-off: smaller values explore distant regions to capture long-range dependencies, while larger values focus on the neighborhood of the starting node.

The transition probability matrix $\mathbf{P}$ is the column-normalized adjacency matrix of the subgraph G_{sub}, where each element

$$P_{ij} = A_{ij} \Big/ \sum_k A_{kj}. \tag{9}$$

This ensures that the sum of transition probabilities from any node equals 1. The elements A_{ij} of the adjacency matrix may be derived from relational weights or simply set to 0 or 1.

For a target category c mapped to node v_c on the knowledge graph, we first calculate the random walk stationary distribution vector r_c starting from the node v_c via the RWR process. The stationary distribution vector r_c is the solution to the following linear system:

$$r_c = (1 - \alpha)\mathbf{P}^\top r_c + \alpha q_c \tag{10}$$

where $\vec{q}_c$ is the initial state distribution vector, implemented as a one-hot vector with a value of 1 at the position corresponding to node c and 0 elsewhere. The notation $\mathbf{P}^\top$ represents the transpose of matrix $\mathbf{P}$. The jth component $r_c(j)$ of vector r_c quantifies the relevance score of reaching node v_j from the starting node v_c while considering the global graph structure. This score assigns an importance value to each node in the graph relative to the starting node v_c.

By computing the RWR vector for every node as the starting point, we obtain the original asymmetric node-node association matrix $\mathbf{R}$

$$\mathbf{R} = \begin{bmatrix} r_1(1) & r_1(2) & \dots & r_1(N) \\ r_2(1) & r_2(2) & \dots & r_2(N) \\ \vdots & \vdots & \ddots & \vdots \\ r_N(1) & r_N(2) & \dots & r_N(N) \end{bmatrix} \tag{11}$$

where element $R_{ji} = r_j(i)$ represents the global relevance score from node v_i to v_j. The matrix asymmetry reflects directional relationships.

For scene context modeling in object detection, we focus on undirected semantic consistency between entities. We define a symmetric semantic consistency matrix S by computing the geometric mean of bidirectional relevance scores for each node pair (v_i, v_j)

$$S_{ij} = \sqrt{R_{ij} \cdot R_{ji}} = \sqrt{r_i(j) \cdot r_j(i)}. \tag{12}$$

This symmetric matrix quantifies the association strength between categories i and j in semantic space with elements $S_{ij} \in [0, 1]$. Excluding background classes, its row/column indices correspond to foreground categories. Higher values indicate a greater likelihood of co-occurrence or functional/attribute correlations in real-world scenarios.

It is crucial to note that the execution of the RWR algorithm is an offline preprocessing step. The semantic association matrix $\mathbf{S}$, as defined subsequently, is computed once prior to model training and stored. During inference, the model directly loads the precomputed matrix $\mathbf{S}$, thereby incurring no iterative graph traversal or random walk computation cost at runtime. This design ensures that the knowledge integration introduces minimal latency during online detection. During inference, it interacts with RT-DETR's category predictions to correct purely visual judgment errors through semantic reasoning, thereby enhancing final detection accuracy and robustness.

As described in Algorithm 1, this process involves calculating the stationary distribution for each node. The time complexity for a single RWR iteration is $O(|E|)$, where $|E|$ is the number of edges in the extracted subgraph. Assuming we set a fixed maximum number of iterations MaxIters = 1000, the total time complexity in the worst case scenario (where the algorithm does not converge early) is the product of the complexity of a single iteration and the maximum number of iterations, i.e., $O(1000 \cdot |E|)$, and this preprocessing is bounded, one-time and does not affect online inference latency.

D. TOKI Module

As illustrated in Fig. 4, to inject external prior knowledge into the detection process, we design a novel TOKI module

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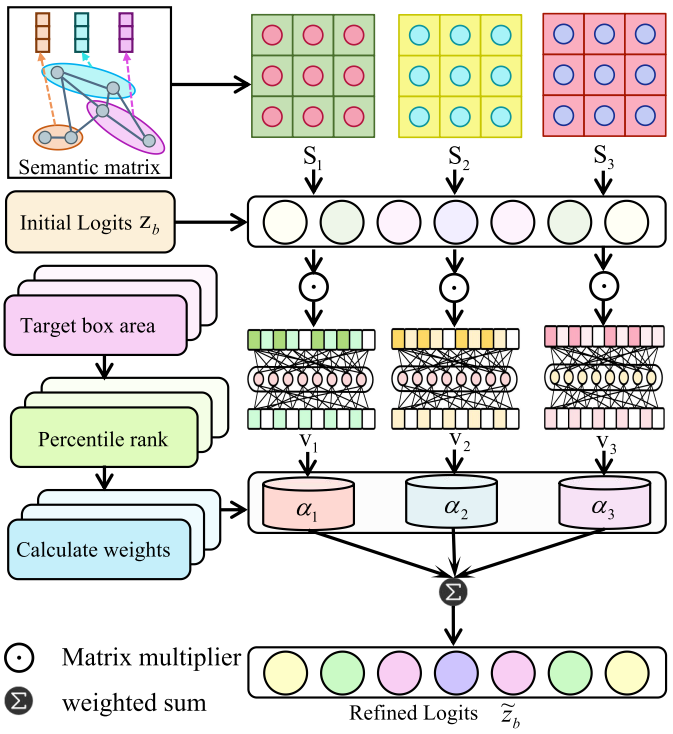


Fig. 4. TOKI module leverages triple-order knowledge representations and dynamic weighted fusion to achieve semantic calibration of the classification logic.

integrated into the model framework. This module leverages structured semantic information from the knowledge graph to recalibrate and enhance initial classification confidence in semantic space, thereby correcting misjudgments caused by visual ambiguity, occlusion, or scale variations while improving discriminative capability.

After the model head processes the input image, for the bth prediction box, it outputs an original class logit vector

$$z_b = W_c \cdot f_b + b_c \tag{13}$$

where f_b is the deep feature vector from the decoder corresponding to the bth prediction box, W_c and b_c are the weight matrix and bias vector of the classification head, respectively, and C is the total number of target categories. The vector z_b contains original judgment scores for target categories based on visual features, ranging over $(-\infty, +\infty)$.

The core innovation lies in fusing this visual prediction score z_b with the prebuilt knowledge graph. Our fusion mechanism transcends simple linear transformation by introducing a data-driven, region-aware multilevel reasoning process.

We first define the first-order semantic affinity matrix as

$$S_1 = S. \tag{14}$$

Which captures direct explicit semantic relationships between categories. the second-order semantic association matrix S_2 is then computed as

$$S_2 = S_1 S_1^T \tag{15}$$

where elements S_{ij} represent semantic affinity between categories i and j through intermediate nodes, capturing longer-range indirect relationships in the knowledge graph.The information encapsulated in S_2 is inferred through S_1, representing a more abstract semantic relationship. This high-order, scene-level information enables the model to encode and share context. For example, although cars and ships may not have a strong direct association in S_1, they may both be associated through the intermediate node of the port, which helps to further enhance the final detection effect based on the first-order knowledge of S_1.

S_2 and S_1 are complementary. S_2 enriches the semantic priors of the model from simple pairwise relationships to a more comprehensive semantic field or contextual embedding, which is particularly crucial for adaptive fusion strategies, where small, information-scarce targets rely more heavily on such second-order reasoning.

The normalized semantic influence matrix is obtained via row-wise Softmax

$$S_3 = \text{Softmax}(S_1). \tag{16}$$

Each row of $\mathbf{S}_3$ forms a probability distribution, representing the normalized distribution of semantic influence across all other categories given the current category.

By multiplying the logit vector z_b with the three semantic matrices, we obtain knowledge-enhanced representation vectors

$$v_1 = S_1 z_b \tag{17}$$

$$v_2 = S_2 z_b \tag{18}$$

$$v_3 = S_3 z_b \tag{19}$$

where v_1 reflects category support based on direct semantic relationships, v_2 captures indirect semantic associations, and v_3 represents category support under normalized semantic influence.

The online computational cost of introducing knowledge fusion into the TOKI module is minimal and deterministic. The core operation involves the matrix-vector multiplications $v_k = S_k z_b$ (for $k = 1, 2, 3$) and the subsequent weighted summation. Given that the semantic matrices S_1, S_2, S_3 are precomputed $C \times C$ matrices, where C is the number of foreground object categories (a small constant), the time complexity for this knowledge fusion stage is $O(C^2)$ per proposal. This is negligible compared to the $O(HW)$ complexity of the visual backbone feature extraction, where H and W are the spatial dimensions of the feature maps. Therefore, the TOKI module maintains high efficiency, ensuring that the integration of structured knowledge does not compromise the real-time inference capability of the overall detection framework.

To adaptively fuse these three-level representations, we propose a weight allocation mechanism based on target relative scale. Denoting the area of the current target box as A_b, and the set of areas from all N proposal boxes in the image as A_b, the percentile rank ρ_b of the current target is defined as

$$\rho_b = \frac{\text{rank}(A_b)}{N} \tag{20}$$

where $\text{rank}(A_b)$ is the ascending rank of A_b in the area set. The percentile rank provides relative scale information of the target in the current context, making no assumptions

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about data distribution and exhibiting strong robustness. Based on ρ_b, the initial weights for the three-level representations are computed as

$$\omega_1 = 1 + \rho_b \tag{21}$$

$$\omega_2 = 2 - \rho_b \tag{22}$$

$$\omega_3 = \rho_b. \tag{23}$$

The above weight function form is relatively simple and intuitive, and is a preliminary design of the method. Its primary goal is to verify the core idea of dynamically adjusting the importance of knowledge of different orders based on the target scale. This design mainly considers two factors: first, strong interpretability, linear correlation between weights and area percentiles, which is easy to understand; Second, it has high computational efficiency, almost no additional computational costs, and serves as a prototype design.

The weighting strategy is theoretically grounded as a means of compensating for information bottlenecks. Larger targets, possessing sufficient visual cues, prioritize first-order ontological knowledge. In contrast, smaller targets suffer from an information deficit, thus requiring a greater reliance on second-order contextual reasoning to resolve ambiguity. Normalized knowledge provides a consistent baseline, with its weights regularized to be less dependent on the target scale.

The initial weights are normalized as

$$\alpha_k = \frac{\omega_k}{\sum_{i=1}^{3} \omega_i}, \quad k = 1, 2, 3. \tag{24}$$

The refined logit vector is obtained through weighted summation

$$\tilde{z}_b = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3. \tag{25}$$

From an interpretability perspective, the above process can be understood as an explicit semantic calibration of the initial visual logits rather than an opaque hidden transformation. Specifically, the semantic matrices interact with the original visual prediction vector z_b to generate structured correction terms v_1, v_2, and v_3, which are then adaptively fused into the refined prediction $\tilde{z}_b$. Therefore, the influence of semantic knowledge on the final decision can be traced through the change from the initial logits z_b to the refined logits $\tilde{z}_b$.

This refined vector $\tilde{z}_b$ serves as the module output: during training, it passes through Softmax to compute cross-entropy loss for model optimization; during inference, it is converted to a probability distribution via Softmax, with final category prediction obtained through argmax. Therefore, TOKI should be understood as a semantic refinement module rather than an independent decision-maker. Its role is to adjust the initial visual classification results according to category-level semantic consistency.

IV. Experimental Results

A. Experimental Dataset

Remote sensing object detection datasets exhibit fundamental differences from natural image datasets, characterized by high object density, a significant proportion of small objects, and frequent occlusions [54]. The considerable variation in spatial resolution, stemming from differing satellite orbits and sensors, leads to large pixel-scale disparities among objects of the same category. Furthermore, complex background clutter and anisotropic object deformation caused by the top-down imaging perspective collectively contribute to substantial intraclass appearance variations, increasing detection complexity.

The DIOR dataset [55] serves as a large-scale benchmark encompassing 20 object categories. It contains a total of 23 463 images, which are divided into 5862 for training, 5863 for validation, and 11 738 for testing, together containing 192 472 annotated instances. All images are standardized at 800×800 pixels, with spatial resolutions ranging from 0.5 to 30 m. Its large scale, complex scenes, and significant scale diversity establish it as a robust benchmark for evaluating algorithm performance, particularly for small object detection.

The NWPU VHR-10 dataset [56] is a widely adopted benchmark for medium-to-high-resolution remote sensing imagery, featuring ten object categories and 3651 annotated instances. Its 650 very high-resolution images are sourced from Google Earth and the Vaihingen dataset. To ensure statistical reliability, we partitioned the 650 annotated images into training, validation, and test sets in a 60:20:20 ratio, resulting in 390, 130, and 130 images, respectively. This dataset provides a reliable platform for validating core detection performance and generalization across diverse domains.

The SIMD dataset [57] is specialized for vehicle detection. It comprises 5000 high-resolution images, each with a resolution of 1024 by 768 pixels. These images contain 45 096 finely annotated instances spanning across 15 distinct categories. It presents a significant challenge due to its abundance of small and medium-sized targets. The dataset is formally divided into 4000 training images, 500 validation images, and 500 test images, facilitating standardized evaluation of model capability in dense small object and fine-grained recognition.

As illustrated in Fig. 5, these three datasets form a complementary multidimensional evaluation framework. DIOR tests robustness in complex, multiscale scenes. NWPU VHR-10 assesses fundamental detection accuracy and cross-domain generalization. SIMD rigorously challenges capabilities in recalling small objects and fine-grained classification. Their high interclass similarity and intraclass diversity are particularly suited for verifying the enhancement of feature discriminability through knowledge graphs.

B. Implementation Details

All experiments were conducted under identical hardware and software configurations. The system was equipped with an Intel Xeon Platinum 8336C CPU and three NVIDIA RTX A6000 GPUs (each with 48 ground breaking (GB) memory), providing sufficient computational resources for processing large-scale remote sensing images. The platform ran on Linux Ubuntu 5.15.0 (64-bit), with all algorithms implemented using PyTorch 2.2.2 and Python 3.9.19.

Extract 2-hop neighbors from ConceptNet 5.5 English version using the class names DIOR, NWPU VHR-10, and SIMD as seeds to construct subgraphs (breadth-first search). The parameter settings for restart probability c, convergence

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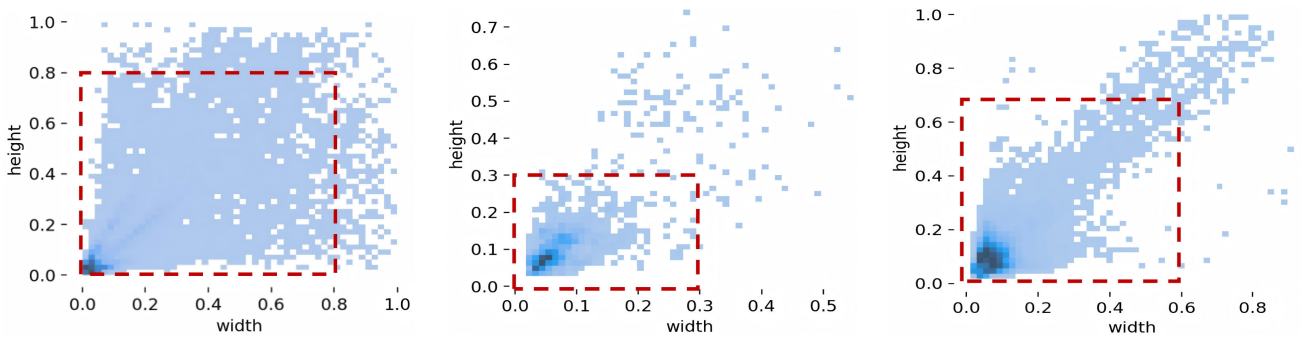


Fig. 5. Distribution of object sizes across DIOR, NWPU VHR-10, SIMD. (a) DIOR. (b) NWPU VHR-10. (c) SIMD.

TABLE I
EXPERIMENTAL PARAMETER SETTINGS

Parameter	Value
Image Size	640 × 640
Number of Worker Threads	8
Optimizer	AdamW
IflrAuto	True
Initial Learning Rate	0.001
Patience	30
Restart probability c	0.15
Epsilon	1e-9
MaxIters	1000

tolerance epsilon, and maximum number of iterations maxIters are shown in Table I.

To ensure fair comparison, we established unified experimental standards while adapting parameters to dataset characteristics. For DIOR and NWPU VHR-10, we used a batch size of 24 and trained for 300 epochs. For the small-object-dense SIMD dataset, we increased the batch size to 32 and extended training to 400 epochs. All images were resized to 640 × 640 pixels, with a consistent learning rate of 0.001. Model performance was evaluated using mean average precision, which comprehensively reflects detection accuracy across categories. Key parameter settings are summarized in Table I.

C. Comparative Experiment

The experiments compared KARFDet against a wide range of mainstream object detection models, including Faster R-CNN [10], RetinaNet [31], the YOLO series with its recent YOLOv12 [62] and YOLOv13 [63] versions, and the DETR series. Our proposed KARFDet model achieved the most superior comprehensive performance across all three datasets.

As illustrated in Table II, on the DIOR dataset, KARFDet attained a mAP_{50} of 65.6%, surpassing all other methods. Notably, it outperformed recent state-of-the-art models such as YOLOv12 with 65.1% and YOLOv13 with 64.6%, as well as the baseline RT-DETR with 63.6%. As illustrated in Table III, this advantage was even more pronounced on the NWPU VHR-10 dataset, where KARFDet significantly outperformed all comparative methods with a mAP_{50} of 91.9%, exceeding the latest YOLOv12 with 85.8% and YOLOv13 with 87.8% by a substantial margin, and demonstrating its strong potential in processing high-resolution remote sensing images with complex ground object scenes. Similarly, as illustrated in Table IV, on the SIMD dataset, KARFDet also ranked first with a mAP_{50} of 75.8%, outperforming all other methods, which confirms its robust generalization capability across different scenario characteristics.

On the DIOR dataset, from a fine-grained category perspective, KARFDet achieved optimal results in multiple key categories, including basketball court (BC), bridge (B), overpass (OP), ship (S), and vehicle (V), with particularly notable advantages in many small or structurally complex object categories. The exceptional performance in key categories such as Airplane, Bridge, and Vehicle on the NWPU VHR-10 dataset demonstrated the model's precise recognition capability for typical remote sensing targets. The experimental results on the SIMD dataset further validated KARFDet's generalization capability, with accurate detection in categories such as Longvehicle, Bus, and Pushbacktruck (PT). These results collectively highlight the advantage of the knowledge graph in understanding spatial relationships between objects, enabling the model to better utilize contextual information for reasoning and judgment. Although performance trade-offs existed in a very few specific categories, our KARFDet model achieved above-average results across all categories, with comprehensive performance reaching state-of-the-art levels, providing an effective solution for object detection in complex remote sensing scenarios.

D. Performance Experiment

As shown in Table V, the comparative experimental results of model performance on the DIOR dataset demonstrate that our proposed KARFDet model achieves an optimal balance between precision, parameter efficiency, and computational complexity. In terms of detection accuracy, KARFDet significantly outperforms all comparative models with an mAP_{50} of 65.6%. More importantly, this performance improvement is achieved while introducing only minimal additional parameters and computational overhead. Specifically, both the parameter count and computational load of KARFDet remain comparable to those of the highly efficient RT-DETR, while being substantially lower than those of other models

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TABLE II

COMPARISON WITH OTHER REPRODUCED METHODS ON DIOR. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	AL	AT	BF	BC	B	C	D	ESA	ETS	GF	GB	HB	OP	S	SD	ST	TC	TS	V	WM	mAP_{50}
Fast RCNN [30]	44.2	66.8	67.0	60.5	15.6	72.3	**52.0**	65.9	44.8	72.1	62.9	46.2	38.0	32.1	**71.0**	35.0	58.3	37.9	19.2	38.1	50.0
Faster RCNN [10]	50.3	62.6	66.0	80.9	28.8	68.2	47.3	58.5	48.1	60.4	67.0	43.9	46.9	58.5	52.4	42.4	79.5	48.0	34.8	65.4	55.5
HSF-Net [58]	51.0	62.1	60.2	80.7	27.4	72.8	43.8	52.1	45.2	67.5	58.7	34.0	48.9	70.8	45.5	53.5	72.9	46.8	37.4	67.0	54.9
CornerNet [59]	64.0	66.3	65.7	**86.3**	34.8	73.1	41.1	60.8	54.2	73.0	66.0	45.3	53.3	81.3	53.5	63.7	80.9	44.1	46.3	**78.8**	61.6
DETR [36]	63.8	**78.6**	71.6	85.1	21.7	**76.3**	41.7	**68.3**	45.4	74.0	**74.2**	24.8	46.1	33.8	36.9	40.0	81.6	47.5	38.3	**78.8**	56.4
DAB-DETR [60]	38.0	57.7	62.5	72.0	22.1	72.2	43.1	53.5	44.4	62.3	48.9	42.3	43.9	12.0	44.0	17.0	58.9	47.0	13.4	45.1	45.0
Deformable DETR [61]	70.9	55.8	65.0	82.7	23.4	75.9	37.1	37.8	45.4	55.1	56.2	9.7	44.9	55.8	38.0	44.4	79.7	36.6	36.1	69.5	51.0
YOLOv3 [13]	72.2	29.2	74.0	78.6	31.2	69.7	26.9	48.6	54.5	31.1	61.1	44.9	49.7	87.4	70.6	68.7	**87.3**	29.4	48.3	78.7	57.1
YOLO-v10 [26]	75.3	66.0	70.1	85.8	27.1	72.5	37.3	46.2	47.7	69.8	63.0	45.7	49.2	86.6	44.0	75.1	84.1	43.8	52.6	69.7	60.6
YOLO-v12 [62]	**76.3**	71.4	**76.9**	83.6	31.4	74.0	49.9	61.5	52.5	73.3	62.3	**63.0**	54.1	87.3	67.0	66.8	86.1	47.7	43.5	73.8	65.1
YOLO-v13 [63]	74.4	71.9	75.1	82.3	31.4	73.1	47.1	59.7	53.5	**74.7**	64.7	62.0	54.3	86.9	64.0	67.2	85.5	48.1	43.1	72.5	64.6
RT-DETR [17]	75.7	64.1	71.1	85.7	34.9	73.4	41.3	54.1	54.0	70.2	67.3	56.4	52.3	88.2	48.7	75.1	84.4	46.3	53.6	74.5	63.6
KARFDet (Ours)	76.1	69.1	71.8	**86.3**	**39.0**	72.9	47.8	55.2	**54.6**	70.3	69.0	58.9	**55.5**	**89.2**	53.4	**76.5**	85.8	**49.7**	**56.5**	74.8	**65.6**

TABLE III

COMPARISON WITH OTHER REPRODUCED METHODS ON NWPU VHR-10. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Airplane	Ship	ST	BD	TC	BC	GTF	Harbor	Bridge	Vehicle	mAP_{50}
Faster R-CNN [10]	82.8	77.5	52.5	96.3	62.9	68.8	98.4	82.5	78.8	63.8	76.4
SSD [28]	95.7	83.1	85.9	96.7	82.3	85.9	57.9	54.6	42.1	75.4	76.0
RetinaNet [31]	98.5	77.5	95.2	96.7	72.1	73.9	99.2	73.3	63.6	68.1	81.8
Soft-NMS [64]	88.3	80.3	60.1	90.4	80.6	89.7	98.9	90.1	67.3	77.6	82.3
Deformable R-FCN [65]	87.3	81.4	63.6	90.4	81.6	74.1	90.3	75.3	71.4	75.5	79.1
M-RCNN [66]	93.2	75.5	92.9	90.4	90.3	**91.2**	95.2	75.2	60.6	74.2	83.9
Deformable DETR [61]	95.7	82.7	54.2	90.4	79.7	83.6	91.6	80.1	67.4	67.6	79.3
DN-DETR [67]	90.2	85.6	89.6	92.4	84.3	67.8	94.7	77.5	73.2	81.3	83.7
YOLOv3 [13]	96.3	87.7	85.7	92.3	81.5	64.4	98.6	82.4	65.6	78.6	83.2
YOLOv10 [26]	94.6	82.8	94.7	94.5	80.2	65.8	97.7	86.7	75.0	74.4	84.7
YOLO-v12 [62]	99.1	**94.3**	94.0	98.2	86.9	59.3	99.3	93.1	58.2	75.9	85.8
YOLO-v13 [63]	98.7	91.8	86.7	**98.6**	90.0	64.2	95.5	**95.0**	68.4	**88.9**	87.8
FFCA-YOLO [68]	99.3	82.1	95.2	97.0	92.1	72.8	**99.6**	93.2	72.1	82.1	88.2
DMEF-F-Net [69]	**99.5**	84.9	**95.8**	95.4	**96.0**	73.2	99.3	94.7	73.4	72.0	88.4
RT-DETR [17]	96.2	90.8	95.6	92.2	95.0	80.9	91.8	93.4	75.3	82.6	89.4
KARFDet (Ours)	99.4	93.4	92.4	96.5	91.7	73.9	95.7	94.2	**93.4**	88.2	**91.9**

TABLE IV

COMPARISON WITH OTHER REPRODUCED METHODS ON SIMD. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Car	Truck	Van	Longvehicle	Bus	Airline	Propeller	Trainer	Chartered	Fighter	Other	Stairtruck	PT	Helicopter	Boat	mAP_{50}
RetinaNet [31]	86.5	67.9	72.4	70.1	79.7	97.2	90.2	91.8	**95.5**	**87.9**	10.6	40.4	10.4	55.8	94.0	70.0
YOLT [70]	84.5	60.6	73.5	59.9	49.0	73.3	66.7	50.0	73.7	0.91	19.1	52.2	21.5	17.4	88.1	52.7
SSD [28]	81.4	64.4	71.5	68.6	78.0	95.3	92.0	90.9	90.5	84.2	14.9	**64.4**	9.1	66.1	87.8	68.8
TOOD [71]	63.7	58.9	57.2	56.7	67.9	83.0	78.4	78.5	78.9	76.5	10.9	24.1	15.2	30.8	75.5	74.3
Deformable R-FCN [65]	91.2	73.2	80.2	81.1	88.3	91.7	92.9	**96.9**	88.1	22.3	25.4	41.1	43.2	41.6	95.8	70.2
YOLOv3 [13]	90.6	75.5	77.5	81.3	88.6	93.0	89.7	88.9	93.6	61.8	22.7	45.9	48.7	33.9	95.5	72.5
YOLOv10 [26]	92.0	74.1	81.0	78.5	86.4	93.0	88.6	93.4	89.7	50.0	25.7	44.2	38.7	66.5	95.7	73.2
YOLO-v12 [62]	**93.5**	**79.9**	**84.9**	79.0	89.9	**97.7**	**99.5**	95.4	92.8	66.5	24.0	41.2	13.5	53.6	**97.3**	73.9
YOLO-v13 [63]	92.5	75.8	82.1	77.4	85.6	97.5	**99.5**	93.2	94.1	78.9	24.0	32.1	18.9	**80.2**	96.3	75.2
RT-DETR [17]	90.6	75.0	78.8	79.8	87.4	91.2	84.6	93.9	89.3	86.3	25.6	33.4	51.2	49.5	95.6	74.2
KARFDet (Ours)	91.4	78.4	80.2	**85.5**	**90.7**	93.6	83.9	93.5	94.4	66.5	**26.6**	47.0	**59.7**	49.6	96.4	**75.8**

with similar accuracy levels. KARFDet successfully achieves a significant leap in detection precision without substantially increasing model complexity. Its high-precision and high-efficiency characteristics endow it with prominent practical value and competitive advantages in remote sensing application scenarios with constrained computational resources.

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TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS

Method	mAP_{50}	Params	GFLOPs
Faster R-CNN [10]	55.5	41.2	198.6
DETR [36]	56.4	54.7M	151.7
DAB-DETR [60]	45.0	44.9M	202.1
Deformable DETR [61]	51.0	39.8M	180.0
YOLOv3 [13]	57.1	63M	245.8
YOLOv10 [26]	60.6	**29.5M**	160.4
RT DETR [17]	<u>63.6</u>	<u>32.7M</u>	**104.9**
KARFDet (Ours)	**65.6**	32.9M	<u>106.1</u>

TABLE VI
COMPARISON OF TIME CONSUMPTION IN EACH STAGE AND FPS

Method	Preprocess	Inference	Postprocess	FPS
RT DETR [17]	5.1ms	17.8ms	0.3ms	43.1
KARFDet (Ours)	1.9ms	12.4ms	50.7ms	15.4

In addition to the overall performance analysis mentioned above, a separate analysis of the time overhead of the KGSE module is conducted. The most computationally intensive step of the KGSE module is the offline generation of the semantic matrix S, which is the random restart walk (RWR) algorithm. We recorded the computation time on three different aerial detection datasets (DIOR, NWPU VHR-10, and SIMD) based on the ConceptNet 5.5 knowledge base. Under our hardware conditions, the maximum knowledge subgraph preprocessing time built on DIOR is 37.87 s. The time on the NWPU VHR-10 dataset is 24.02 s, and the time on the SIMD dataset is 27.12 s. This indicates that the entire knowledge graph encoding process that forms the semantic foundation of our model can be completed in less than a minute on a standard machine. Importantly, this is a one-time cost before the model training begins and does not increase the time for online inference.

As shown in Table VI, we benchmark the inference speed of KARFDet and baseline RT-DETR under the same GPU hardware with consistent input resolution and batch size. RT-DETR achieves an FPS of 43.1, with 5.1 ms preprocessing, 17.8 ms inference, and 0.3 ms postprocessing per image. KARFDet achieves faster preprocessing (1.9 ms) and inference (12.4 ms) speeds, but its overall FPS decreases to 15.4 due to increased postprocessing overhead (50.7 ms) introduced by the knowledge-guided refinement steps. It is worth emphasizing that the KGSE module operates entirely offline and does not contribute to inference-time overhead. The additional postprocessing time reflects the additional online knowledge retrieval operations performed by the TOKI module. Notably, the GFLOPs of KARFDet remain nearly identical to those of the baseline (106.1 versus 104.9), indicating that the model does not introduce additional floating-point computations. The FPS decrease is primarily attributed to memory-bound operations, such as reading the semantic knowledge base, tensor copying, and dimension transformations, rather than an increase in computational workload. Consequently, on hardware platforms with higher memory bandwidth or optimized memory access patterns, the FPS of KARFDet is expected to improve further. This trade-off is justified by a substantial accuracy gain of +3.1% mAP@50 on DIOR, demonstrating that KARFDet achieves a favorable balance between detection precision and inference efficiency.

To evaluate the generalization ability of the model under different data modalities, we conducted cross-sensor experiments from optical images to infrared images on the VEDAI dataset. The results show that KARFDet mAP@50 is 0.531, slightly lower than the pure visual baseline RT-DETR of 0.564. This result indicates that the effectiveness of current knowledge fusion mechanisms is constrained when facing cross-sensor scenes with vastly different imaging mechanisms and varying underlying feature distributions. This indicates that the current knowledge fusion mechanism is limited under cross-modal conditions. We analyze that the reason for this is that the knowledge semantic calibration relies on feature representations extracted by the visual backbone network that are aligned with semantic concepts. Moreover, because the TOKI module operates on the initial classification logits produced by the detection head, downstream knowledge fusion cannot fully compensate when the upstream visual representations have already been degraded by modality shift. This experiment clarified that KARFDet is better at semantic enhancement and correction of visual targets within the same modality, rather than directly solving the problem of low-level perception failure caused by different imaging principles.

E. Ablation Experiment

The KARFDet framework comprises three core components: the MSFKA module and the KGSE module, with the TOKI module, in which the KGSE module serves as the upstream processor to construct the semantic consistency matrix, an essential prerequisite for the downstream TOKI module to perform its triple-order knowledge fusion. As demonstrated in Tables VII–IX, ablation studies conducted on the DIOR, NWPU VHR-10, and SIMD datasets consistently show that each module independently delivers significant performance improvements, while their combination produces a remarkable synergistic effect.

On DIOR, MSFKA alone boosted detection accuracy for baseball field (BF), goldfield (GF), GB, and Stadium (ST) by enhancing multiscale feature representation through parallel convolutional kernels. Independently, the integrated KGSE+TOKI unit improved performance on airport (AT), basketball court (BC), and train station (TS) by embedding semantic priors and spatial constraints. The full KARFDet model achieved an optimal 65.6% mAP_{50}, demonstrating deep coupling between feature enhancement and knowledge guidance. This bidirectional mechanism excelled in detecting airplane (AL), HB, OP, ahip (S), and vehicle (V). ConceptNet is a large-scale, multilingual knowledge graph with the core advantage of capturing human general cognition of the world, and all target categories in the experimental dataset have links found in this knowledge graph. From the analysis of the ablation experiment results in Tables VII–IX above, it can be seen that using the semantic relationship matrix extracted

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TABLE VII

ABLATION EXPERIMENT ON DIOR. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modules		AL	AT	BF	BC	B	C	D	ESA	ETS	GF	GB	HB	OP	S	SD	ST	TC	TS	V	WM	mAP_{50}
MSFKA	KGSE+TOKI																					
○	○	75.7	64.1	71.1	85.7	34.9	73.4	41.3	54.1	54.0	70.2	67.3	56.4	52.3	88.2	48.7	75.1	84.4	46.3	53.6	74.5	63.6
●	○	74.6	69.2	**72.1**	85.5	37.0	**73.6**	45.8	53.1	**55.4**	**72.6**	**69.4**	55.1	54.4	88.7	**56.0**	73.6	85.3	45.6	54.2	74.4	64.8
○	●	75.2	**70.5**	71.6	**86.3**	36.1	73.5	44.8	54.2	54.9	71.4	67.8	54.9	51.5	88.5	48.1	73.9	84.9	**52.3**	53.9	73.5	64.4
●	●	**76.1**	69.1	71.8	**86.3**	**39.0**	72.9	**47.8**	**55.2**	54.6	70.3	69.0	**58.9**	**55.5**	**89.2**	53.4	**76.5**	**85.8**	49.7	**56.5**	**74.8**	**65.6**

TABLE VIII

ABLATION EXPERIMENT ON NWPU VHR-10. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modules		Airplane	Ship	ST	BD	TC	BC	GTF	Harbor	Bridge	Vehicle	mAP_{50}
MSFKA	KGSE+TOKI											
○	○	96.2	90.8	95.6	92.2	95.0	80.9	91.8	93.4	75.3	82.6	89.4
●	○	99.0	86.9	**95.9**	94.3	93.5	**87.0**	**96.1**	91.3	81.5	83.4	90.9
○	●	98.4	**94.7**	92.4	95.1	**96.0**	77.1	91.8	85.2	88.2	87.1	90.6
●	●	**99.4**	93.4	92.4	**96.5**	91.7	73.9	95.7	**94.2**	**93.4**	**88.2**	**91.9**

TABLE IX

ABLATION EXPERIMENT ON SIMD. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modules		Car	Truck	Van	Longvehicle	Bus	Airline	Propeller	Trainer	Chartered	Fighter	Other	Stairtruck	PT	Helicopter	Boat	mAP_{50}
MSFKA	KGSE+TOKI																
○	○	90.6	75.0	78.8	79.8	87.4	91.2	84.6	93.9	89.3	**86.3**	25.6	33.4	51.2	49.5	95.6	74.2
●	○	**91.5**	74.1	78.9	79.2	88.4	93.1	**92.5**	93.0	91.9	61.8	**28.3**	38.0	57.5	63.1	95.9	75.2
○	●	91.0	74.3	79.8	83.5	87.9	**93.9**	86.6	**95.0**	90.8	66.5	18.7	44.2	47.5	**66.8**	96.1	74.9
●	●	91.4	**78.4**	**80.2**	**85.5**	**90.7**	93.6	83.9	93.5	**94.4**	66.5	26.6	**47.0**	**59.7**	49.6	**96.4**	**75.8**

from ConceptNet for optimization significantly improves performance compared to the version that only uses image feature checking. From the perspective of final task performance, it indirectly but strongly proves that using this knowledge base for enhancement in this article can extract more semantic information that is beneficial and generalizable for remote sensing object detection tasks.

Cross-dataset validation reinforced these findings. On NWPU VHR-10, MSFKA improved the detection of storage tank (ST) and BC, while KGSE+TOKI aided ship and tennis court (TC) recognition. Their integration achieved peaks in the AL and vehicle categories. On SIMD, MSFKA enhanced car and propeller detection, whereas KGSE+TOKI improved airline and long vehicle identification. Combined use yielded synergy in the truck and bus categories. The ablation experiment has demonstrated that the KGSE+TOKI module incorporating this weight strategy as a whole makes a decisive contribution to performance improvement, demonstrating the effectiveness of the weight design in (21), (22), and (23).

In the chimney category (C) of the DIOR dataset, the full model's precision experienced a slight decrease compared to the single-module variants, reflecting the feature optimization process during multimodule collaboration. Similarly, the BC category in the NWPU dataset and the Fighter category in the SIMD dataset also demonstrated scenarios where individual modules achieved superior performance. These observations collectively indicate that KARFDet effectively performs performance balancing during module integration, where its objective is not to achieve peak performance in every individual category but to realize global performance maximization through internal calibration.

In summary, MSFKA acts as a perceptual engine for multiscale features, and KGSE+TOKI as a cognitive engine for knowledge reasoning. Their integration enables KARFDet to unify perception and cognition, advancing remote sensing image interpretation.

F. Error Analysis

Notwithstanding its strong performance, the KARFDet framework has certain limitations. Its effectiveness is contingent on the quality of the underlying knowledge graph; incomplete or noisy relational data can occasionally lead to suboptimal guidance from the TOKI module. In order to alleviate this dependency, three solutions can be proposed in the future. First, before constructing the subgraph G_{sub}, a threshold can be set to remove the relationship type R with extremely weak correlation with the target category set L and reduce the noise edges of the source. Second, before calculating RWR, perform weight pruning on the edges in the knowledge graph to remove low-weight edges and prevent interference from weak correlation noise during random walks. Third, vertical knowledge maps specifically designed for the remote sensing field can be introduced or constructed to ensure the quality of the maps.

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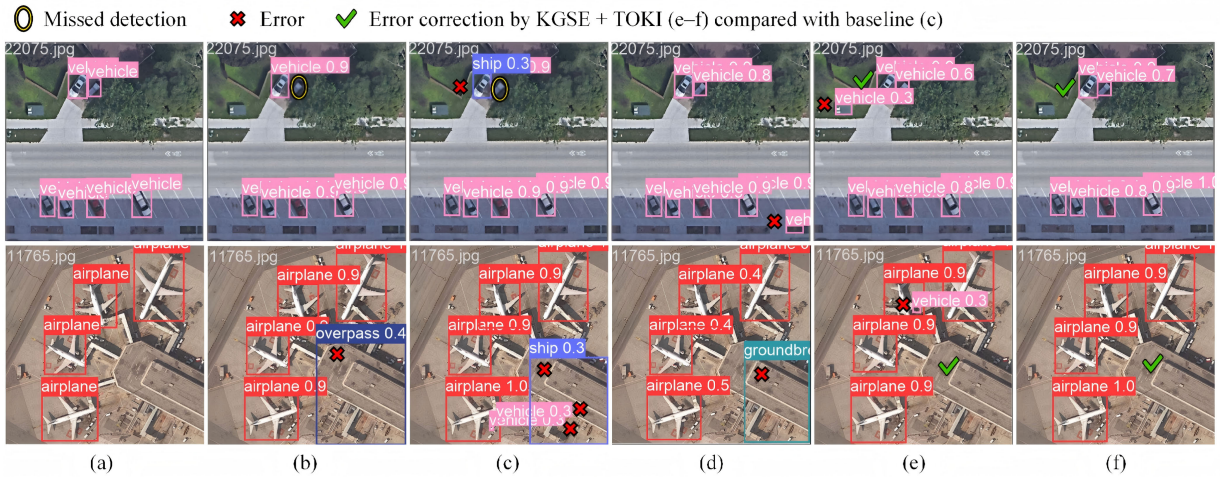


Fig. 6. Detection performance comparison on DIOR. (a) Annotation box, (b) YOLOv10, (c) RT-DETR, (d) RT-DETR+MSFKA, (e) RT-DETR+KGSE+TOKI, (f) KARFDet (Ours).

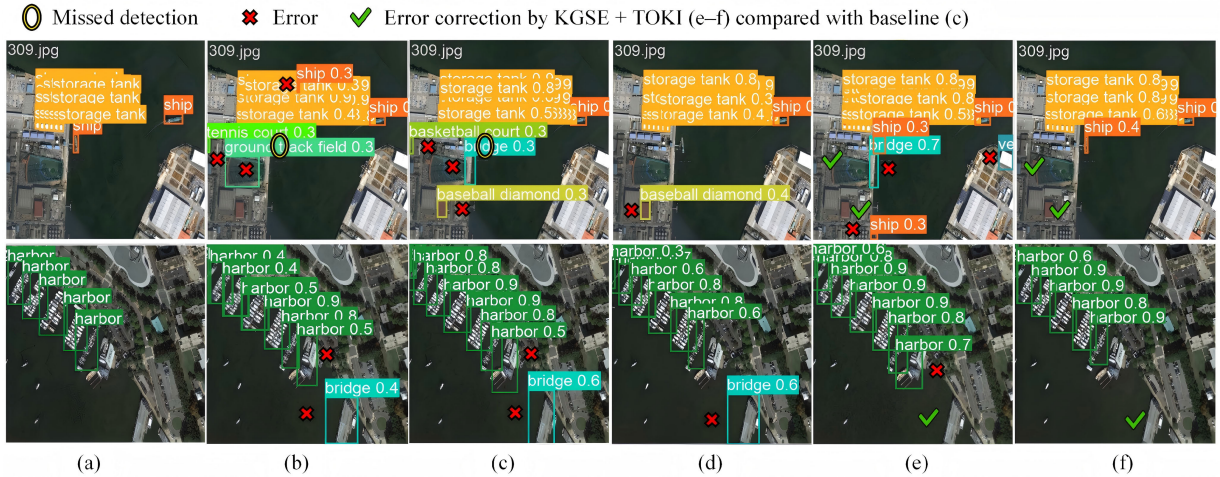


Fig. 7. Detection performance comparison on NWPU VHR-10. (a) Annotation box, (b) YOLOv10, (c) RT-DETR, (d) RT-DETR+MSFKA, (e) RT-DETR+KGSE+TOKI, (f) KARFDet (Ours).

Furthermore, the fusion process between the knowledge graph and visual features operates as a "black box," lacking sufficient interpretability in its decision-making logic. One solution is to judge the fusion process by analyzing the semantic incidence matrix. The semantic incidence matrix provides clear prior knowledge about the relevance of categories, and the refinement from the initial visual logic to the final logic provides a direct way to analyze how semantic guidance enhances or suppresses the response of specific categories.

Additionally, the KARFDet model exhibits a slight increase in both parameter count and computational complexity compared to the baseline RT-DETR. While this trade-off is justified by significant accuracy gains, it suggests an *a*-direction for future work in model streamlining for resource-constrained environments.

G. Visualization Results

Visualization analysis across datasets reveals systematic improvements throughout the model evolution:

Figs. 6–8 provide visual evidence of KARFDet's error correction capabilities. The red crosses mark (×) erroneous detections made by the baseline models, whereas the green check (✓) marks highlight the corresponding cases where KARFDet's semantic reasoning successfully resolves the visual ambiguity.

As illustrated in Fig. 6, YOLOv10 exhibited significant false positives and missed detections in complex backgrounds, while RT-DETR showed limited discrimination capability. The MSFKA module enhanced multiscale target capture but retained false detections due to a lack of semantic understanding. The KGSE+TOKI module suppressed semantically inconsistent errors but missed some targets. Ultimately, KARFDet achieved zero category misclassifications through modular synergy, demonstrating superior anti-interference capability and recognition accuracy.

As illustrated in Fig. 7, both YOLOv10 and RT-DETR showed numerous detection errors. The MSFKA module enhanced detail perception for targets like HBs, while the KGSE+TOKI module eliminated incorrect detections through spatial reasoning. However, as illustrated in the upper part of Fig. 7(e), the ship and bridge are still mistakenly detected even with knowledge graph enhancement. This indicates that

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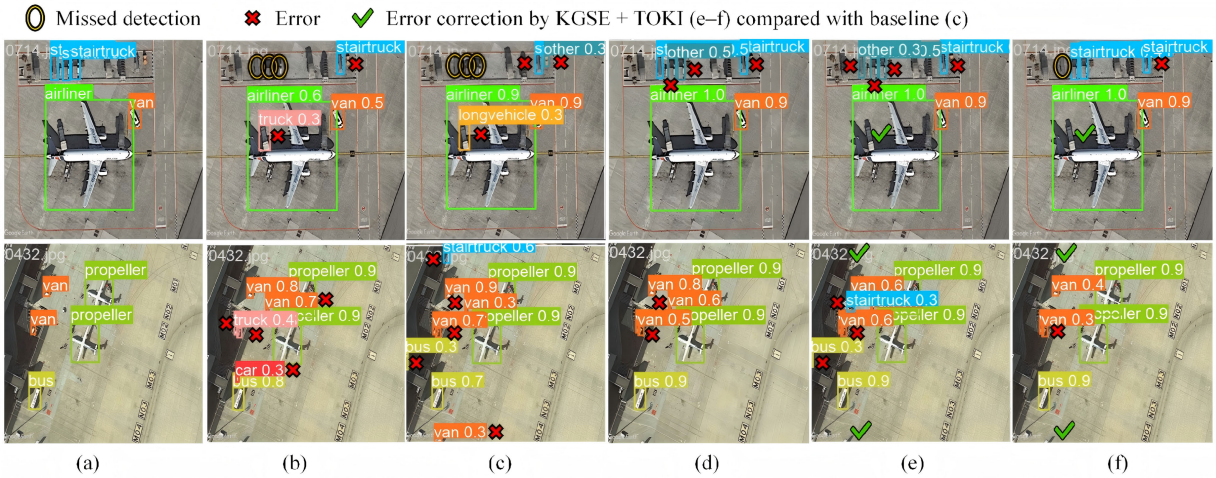


Fig. 8. Detection performance comparison on SIMD. (a) Annotation box, (b) YOLOv10, (c) RT-DETR, (d) RT-DETR+MSFKA, (e) RT-DETR+KGSE+TOKI, (f) KARFDet (Ours).

when a strong semantic relationship exists between a falsely detected object and other objects in the scene, the knowledge graph may consider the detection plausible, making it difficult to eliminate this type of false positive. KARFDet achieved optimal detection completeness with zero missed detections and precise bounding boxes in dense scenarios.

As illustrated in Fig. 8, YOLOv10 and RT-DETR demonstrated insufficient detection completeness with multiple false positives. The MSFKA module improved vehicle recognition, while the KGSE+TOKI module corrected classification errors through logical reasoning. KARFDet showed the most pronounced advantages in distinguishing similar targets within complex scenes.

Comprehensive analysis confirms KARFDet's significant advantages in detection robustness and precision across various challenging scenarios. This stems from the organic integration of the MSFKA module's local feature enhancement and the KGSE+TOKI module's semantic reasoning, achieving comprehensive optimization from feature extraction to contextual understanding.

V. Conclusion

This article proposes KARFDet, a novel remote sensing object detection framework that integrates multiscale visual feature extraction with structured knowledge graph reasoning. To address the characteristic challenges of significant scale variations and complex spatial relationships in aerial imagery, our research achieves technical breakthroughs through a tripartite structure. The MSFKA module enhances multiscale feature representation for targets ranging from large structures to small vehicles. The KGSE module serves as the foundational knowledge base, constructing a semantic consistency matrix that quantifies associative relationships between object categories. Building upon this, the TOKI module performs dynamic, region-adaptive fusion of this multiorder knowledge into the visual detection process. This integrated system enables a cognitive-closed-loop where visual perception and semantic reasoning enhance each other.

This study provides an effective solution for achieving unified perception and reasoning in remote sensing image interpretation. Future research will focus on developing a more adaptive knowledge learning framework, exploring a more refined and learnable weight generation network to enhance the coverage of real scenes, and implementing a lightweight design through model compression technology to promote its deployment in edge computing environments.

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Xin Cheng received the B.S. degree in science and the M.S. degree in engineering from Xi'an University of Posts and Telecommunications, Xi'an, China, in 2012 and 2016, respectively. She is currently pursuing the Ph.D. degree with Wuhan University of Technology, Wuhan, China.

Her research interests include deep learning and computer vision.



Yibo Liu received the B.S. degree in computer science from Xi'an University of Posts and Telecommunications, Xi'an, China, in 2012, and the Ph.D. degree from the National University of Defense Technology, Changsha, China, in 2024.

He is currently a Lecturer with the Information Support Force Engineering University, Wuhan, China, and Hubei Provincial Key Laboratory of Data Intelligence, Wuhan. His research interests include deep learning, computer vision, and knowledge engineering.



Haisu Zhang received the Ph.D. degree in computer applications from the University of Science and Technology, Nanjing, China, in 2012.

He is currently a Professor with the Information Support Force Engineering University, Wuhan, China. His research interests include cross-modality analytics, image processing, and data engineering.



Huan Liu received the M.S. degree in electronic information from Hubei Normal University, Huangshi, China, in 2025. He is currently pursuing the Ph.D. degree in information and communication engineering with the School of Information Engineering, Wuhan University of Technology, Wuhan, China.

His research interests include image processing, machine learning, object detection, and remote sensing.



Yao Wu the B.S. and Ph.D. degrees from the Huazhong University of Science and Technology, Wuhan, China, in 2016 and 2023, respectively.

She is currently an Assistant Professor with the National University of Defense Technology, Changsha, China. Her research primarily lies in data mining and deep learning, with a specific focus on graph data mining and graph neural networks.



Zhiliang Hu received the M.S. degree from Xiamen University of Technology, Xiamen, China, in 2025. He is currently pursuing the Ph.D. degree with the School of Information Engineering with Wuhan University of Technology, Wuhan, China.

His research interests include object detection and deep learning.



Ning Xu (Member, IEEE) received the Ph.D. degree from the University of Electronic Science and Technology of China, Chengdu, China, in 2003.

He was a Visiting Scholar at Texas A&M University, College Station, TX, USA, and is currently a Professor with the School of Information Engineering with Wuhan University of Technology, Wuhan, China. He has published over 80 research articles. His research interests include image processing and artificial intelligence.

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